

Protocol Analysis
Experiment #1
Protocols : HTTP, DNS & TCP

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Application Scenario

Client requests a file from a server

Server returns the file

Nature of the file : html : Simple one line text

Protocols to be analysed

Analyse how protocols DNS, HTTP & TCP are used in running this application. Analysis to be done in 3 parts

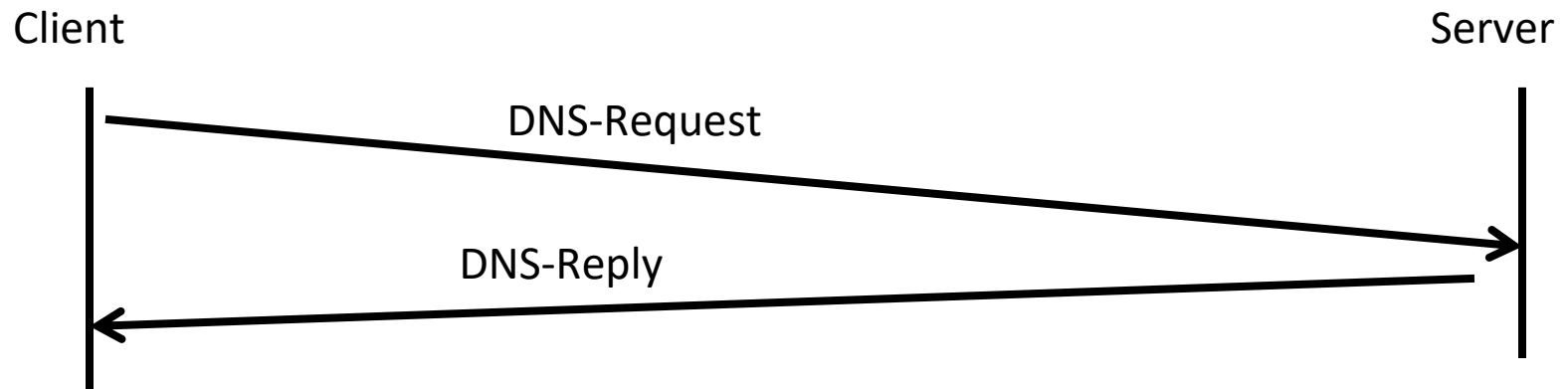
Part#1 : DNS

Part #2 : HTTP

Part #3 : TCP

Trace File to be analysed : HTTP-Tracefiles\http-trace-1

Part#1 : Protocol - DNS



- Start Wireshark; Open the trace file ; Apply DNS in the filter window
- Analyse

Write down the following

1. IP address of the client ?

Ans :

2. IP address of the DNS

Ans :

DNS query

1. Source port in UDP

Ans :

2. Destination port in UDP

Ans :

3. Is DNS search recursive or iterative (Check the flags)

Ans :

4. What is the query ?

Ans :

DNS Reply

1. How many Authoritative Resource Records?

Ans :

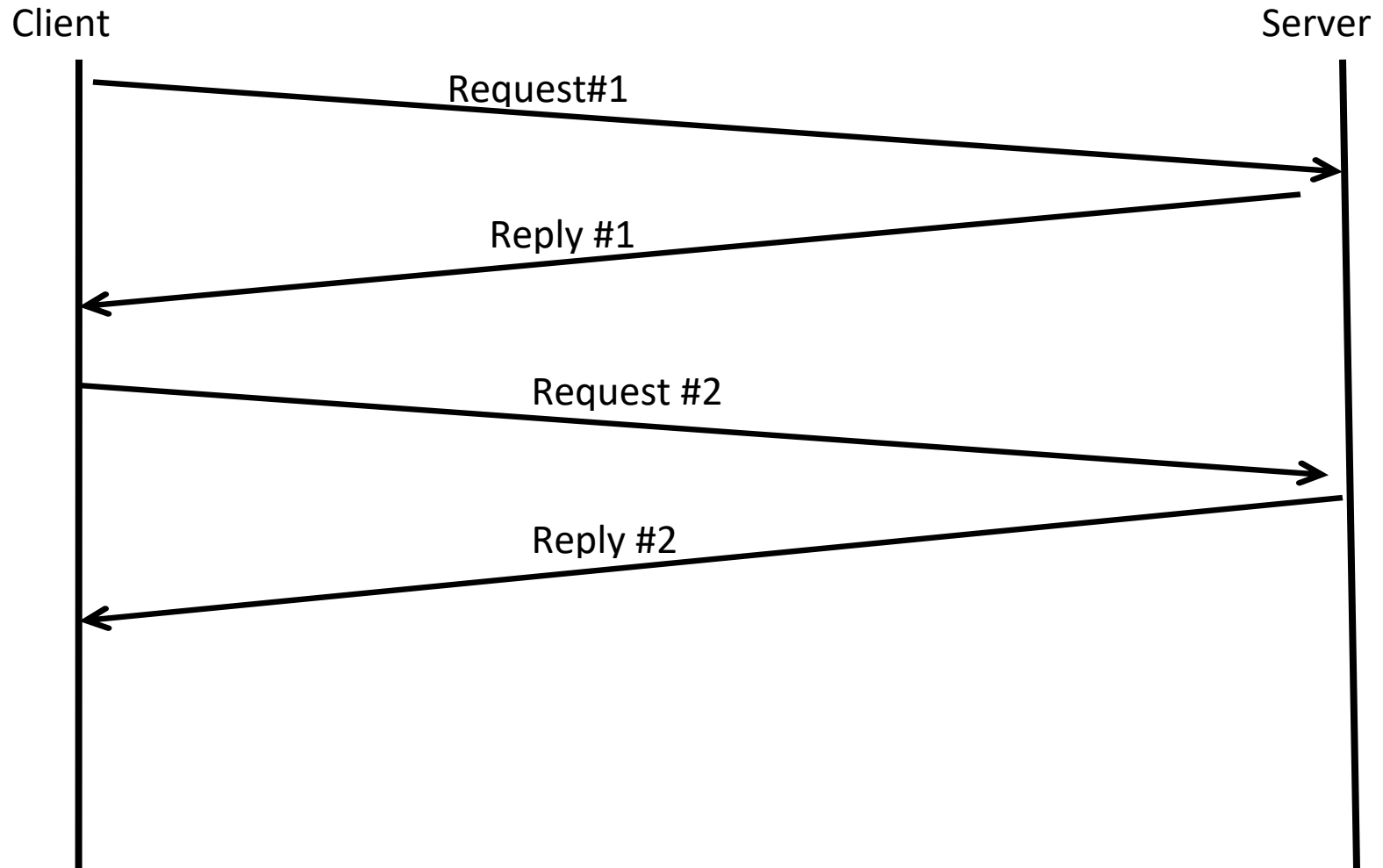
2. How many Additional Resource Records?

Ans :

3. What is the IP address returned in the reply?

Ans :

Protocol : HTTP



Protocol : HTTP

Enter HTTP in the filter, Analyse and answer the questions

Request #1 :

1. What is the browser used in the client?

Ans :

2. What is the name of the server ?

Ans :

3. What is the name of the file requested?

Ans :

4. Is the connection persistent or non-persistent ?

Ans :

Reply #1

1. What is the size of the content returned ?

Ans : Content length : Total length :

2. What is the main text content of the reply ?

Ans : Content length :

Request #2

1. What is the name of the file requested?

Ans :

Reply #2

1. What is the reply in the status?

Ans :

2. What is the size of the contents replied?

Ans : content length :

Total length :

Protocol : TCP

Apply TCP in the filter window

1. Analyse
 1. Connection establishment phase
 2. Data transfer phase
2. Fill up the values for the attributes mentioned in the next slide Primarily Seq&Ack analysis
3. Draw Timeline diagram

Protocol : TCP

Client

IP add :192.168.1.102

Server

IP add : 128.119.245.12

